PORTFOLIO PROJECT PDF

Gun Violence Incident Analysis and High-Casualty Prediction

This is a shorter portfolio version of my Fall 2024 Data Mining final project. I used R to study gun violence incidents in the U.S. and check what patterns were linked to more severe cases. My main work was cleaning the data, building time and location features, parsing participant fields, exploring text patterns, and testing a classification model.

R | EDA | Random Forest | Text Mining | Data Mining

SIMPLE NOTE

This PDF rewrites my original course report, slides, R script, and supporting plots into one cleaner portfolio version for HR and interview reading.

BUSINESS PROBLEM

Use incident-level gun violence data to find clear patterns behind severe incidents and turn a large raw dataset into practical, readable insights for public-safety stakeholders.

PROJECT INFO

At a glance

Course: ALY6040 - Data Mining

Term: Fall 2024

Type: Individual project

Team: N/A

Original deliverables: Report, slides, R script

Dataset: Kaggle Gun Violence Dataset

Rows / columns: 239,677 incidents / 29 fields

Main tools: R, tidyverse, ggplot2, caret, randomForest, tidytext

Model note: High-casualty target = killed + injured >= 3; Random Forest was the main final model I kept.

OVERVIEW

Project Snapshot

The original school project included a long final report, a slide deck, a full R script, and many plots. In this shorter version, I focus on the business goal, the workflow I owned, the strongest visuals, and the final honest read from the model results.

BUSINESS FOCUS

Severity, timing, and hotspots

I studied when incidents happened more often, where they clustered, and what was linked to high-casualty cases.

DATA SCALE

2013 to 2018 U.S. incident data

The dataset covered 239,677 records and gave enough size for trend checks, participant views, and model testing.

METHODS USED

EDA + parsing + text mining + model comparison

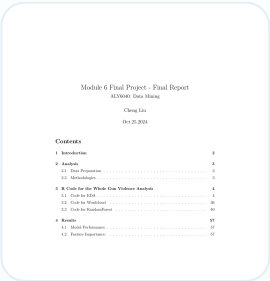
I combined descriptive analysis with feature engineering and a simple classification task.

ORIGINAL DELIVERABLES

Report, slides, code, and plots

This portfolio PDF reuses the core story but makes it much faster to read in one sitting.

Original source materials consolidated here


Module 6 Final Project – Final Report
AI/6040: Data Mining
Cheng Liu
Oct 25 2024
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Course final report

Presentation deck

I kept the main charts from the original work and rewrote the report in simpler English so the project is easier to skim.

My role in the project

- Completed the full project as an individual assignment.
- Cleaned the data and checked missing values, duplicates, and outliers.
- Built date, location, and severity features for analysis and modeling.
- Parsed participant fields and explored location text with word clouds.
- Tested Logistic Regression, KNN, LightGBM, and Random Forest.
- Wrote the report and prepared the presentation materials.

Contribution note

This was my own end-to-end project. I also checked Kaggle notebooks and discussions to improve my approach on messy fields and model setup.

BUSINESS PROBLEM AND ANALYSIS PLAN

Problem, Dataset, and Project Proposal

I framed this project as a public-safety analytics task: first understand the data story well, then test whether incident-level fields could help flag higher-severity cases.

PROJECT PURPOSE

I wanted to turn a large, messy incident dataset into a clear report with patterns that a manager could understand fast. I also wanted to see whether a simple model could separate high-casualty incidents from the rest.

Dataset summary	Details
Source	Kaggle Gun Violence Dataset
Time range	January 2013 to March 2018
Main structure	One incident-level table with 239,677 rows and 29 variables
Working data	Raw incident fields plus model-ready features such as year, quarter, month, weekday, and high_casualty
Main missing issue	Several fields had gaps, especially n_guns_involved and district-related variables

STAKEHOLDERS

Policymakers, law enforcement, and public-safety planners who need a simple read on where and when incidents appear more often.

MAIN QUESTIONS

What factors predict severe incidents?
Are there times of year when incidents are more likely?
Which incidents are more likely to have multiple casualties?
How do local factors relate to high-rate states?

PROJECT PROPOSAL

Load and clean the data, build date and severity features, run EDA, parse participant fields, explore location text, compare a few models, then write a short recommendation summary.

SELECTED CODE - MAIN SETUP

```
gun_data <- read.csv("gun-violence-
data_01-2013_03-2018.csv")

gun_data$date <- ymd(gun_data$date)
gun_data <- gun_data %>%
  mutate(
    year = year(date),
    quarter = quarter(date),
    month = month(date, label = TRUE),
    weekday = wday(date, label = TRUE)
  )

gun_data$total_casualties <- gun_data$n_killed +
gun_data$n_injured
gun_data$high_casualty <- ifelse(gun_data$total_casualties
>= 3, 1, 0)
```

I first turned the raw incident table into a model-ready version with date features and a simple severity target.

LAB / CODE NOTE

I used one main R script for the full analysis and an R Markdown file for the report. The code covered EDA, parsing, text mining, model testing, and plotting.

EXAMPLE FIELDS I USED

state, city_or_county, n_killed, n_injured

participant_age, participant_gender, participant_status, participant_type

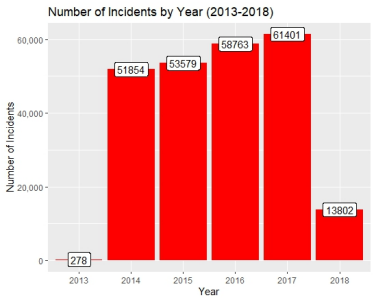
incident_characteristics, location_description, year, month, weekday

EXPLORATORY DATA ANALYSIS

Key Findings and Visual Evidence

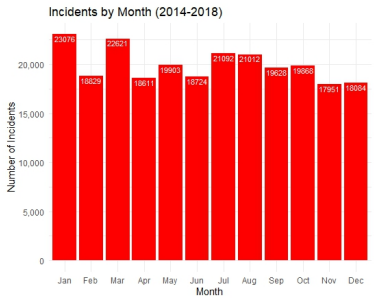
The strongest part of this project was the EDA. Before I even reached the model, the charts already showed a clear story about time patterns and geographic concentration.

Incidents kept rising from 2014 to 2017



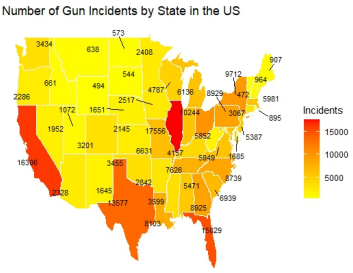
The dataset shows a steady increase from 51,854 incidents in 2014 to 61,401 in 2017. 2018 is partial because it only goes through March.

Month pattern was not flat



January was the highest month in this chart, with March, July, and August also standing out. This suggested a useful seasonal pattern to check further.

Hotspots were clear by state



Illinois, California, Florida, and Texas had the highest incident volumes. Chicago also appeared as the top city in the city-level chart.

WHAT THESE CHARTS TOLD ME

- The incident count increased year by year across most of the full-period data.
- Time mattered: early-year months and summer months showed heavier activity.
- Weekends were also higher in the weekday chart, especially Sunday and Saturday.
- Geography mattered a lot. A few states and cities dominated the incident counts.

Quick business read

In this dataset, gun violence was not evenly distributed. Time and place already gave a strong first filter for where attention should go.

METHODS AND MODEL NOTE

Feature Engineering, Text Mining, and Final Model Read

I did more than simple counting charts. I also parsed multi-value participant fields, explored text in location descriptions, and compared several models for the high-casualty target.

SELECTED CODE SNIPPET

```
gun_data_clean <- gun_data %>% drop_na()

gun_data_encoded <- dummy_cols(
  gun_data_clean,
  select_columns = c("state", "month", "weekday"),
  remove_selected_columns = TRUE
)

rf_model <- randomForest(
  high_casualty ~ . - total_casualties,
  data = train_data_scaled,
  ntree = 100
)
```

The model step used one-hot encoding, a binary high-casualty target, and Random Forest as the final main classifier.

INTERPRETATION

- I created a simple severity target: high_casualty = 1 when killed + injured was 3 or more.
- I tested Logistic Regression, KNN, LightGBM, and Random Forest, then kept Random Forest as the main final model.
- The target class was rare, so accuracy alone looked better than the real minority-class performance.
- The final report showed strong majority-class prediction but weak recall on rare severe cases.

ACCURACY	PRECISION	RECALL	F1 SCORE
97.45%	86.67%	2.94%	5.68%

Final model read

The Random Forest result was honest but limited. It was much stronger at recognizing the common class than catching rare high-casualty incidents.

Location text also added context

apartments
park
school
neighborhood
station
center
village
bar
inn
airport
club
west
gas
lounge
university
county
house
store
east
mart
square
market
community
complex/river
middle
hills
plaza
south
north
lake
church
creek
austin
international
mobile
apartment
walmart
food
grill
home
heights
shop
elementary
manor
central
gardens
district
hotel
bank
eleven
hospital
springs
chatham
regional
country
crossing
exton
viewtwp
woods
harbor
landing
guns
night
casino
marker
union
chicken
auto
san
medical
estates
yards
waffle
island
car
indian
shooting
nightclub
field
town
tavern
express
grocery
forest
office
brighton
pullman
townhomes
western
woodlawn
modonalds
bollegs
police
dfw
villa
housing
sports
garfield
restaurant
dollar
city
pub
taco
family
valley
chicago
beach
modonald
garden
bp
pine
bay
pointe
america
chevron
vistagrandtownship
hall
gresham
terrace
dallas
shore
oaks
pawn
pizza
national
bell
mile
nina
american
trailer
days
corner
shopping
worth
shell
ward
bridge
grove
memorial
washington
academy
convenience
green
marathon
valero
humboldtcitgo
quick
chester
mountain
dorch
bank
run
manor
elementary
homes
hospital
springs
chatham
burger
lodge
range
king

EXECUTIVE SUMMARY

Conclusion, Recommendations, and Interview Use

This project shows how I organize a full analytics workflow: start with a messy dataset, build a clean story with visuals, test a model, and stay honest about the limits of the result.

MAIN FINDINGS

Incident counts rose across the main years in the dataset.

Time and place mattered a lot: some months, weekends, and a few states and cities clearly stood out.

RESULT MEANING

The project was strongest as a pattern-discovery and communication project. The model added value, but the EDA gave the clearest and most stable story.

RECOMMENDATIONS

Use the time and hotspot patterns for targeted planning, and improve the model later with class balancing, richer outside data, and stronger validation.

WHAT THIS PROJECT SHOWS ABOUT MY WORK

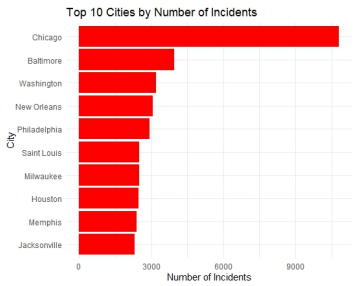
- I can take a large raw dataset and turn it into a structured analysis plan.
- I can work with messy multi-value fields, not only clean textbook data.
- I do not overclaim model quality when class imbalance makes the task hard.
- I can rewrite technical work into a clearer manager-facing summary.

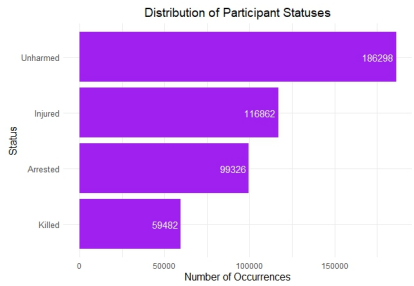
HOW I WOULD EXPLAIN THIS IN AN INTERVIEW

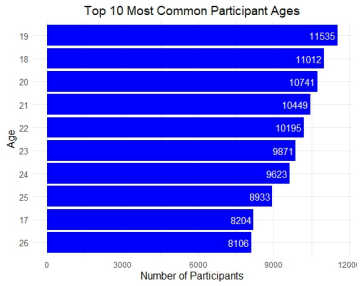
Simple STAR version: I had a large U.S. gun violence dataset with messy fields and many possible directions. My job was to turn it into a clear project and test whether severe incidents could be flagged. I cleaned the data, engineered time and severity features, parsed participant columns, explored text, and compared several models. In the end, I found clear time and location patterns, and I also learned that class imbalance made the prediction task much harder than accuracy alone suggested.

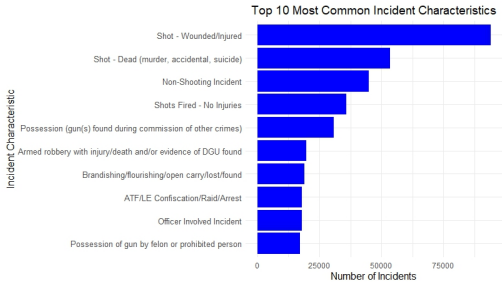
> **Final note**
> This was an individual course project. The strongest part was the full process and the honest interpretation, not pretending the model was stronger than it really was.

Quick Visual Gallery









I kept a small gallery here to show that the project covered location, participant, and incident-type views, not only the model output.